STROKE

SOURCE-WHO

Key facts

- In 2021, stroke was the third leading cause of death and disability globally, with an estimated 93.8 million cases.
- In 2021, there were 11.9 million new cases of stroke.
- The lifetime risk of stroke has increased by 50 % over the past 20 years, with 1 in 4 adults predicted to experience a stroke in their lifetime.
- The global burden of stroke measured in Disability-Adjusted Life Years (DALYs) increased from 137 million in the year 2000 to 160 million in 2021 *(1)*.
- Most of the stroke burden is attributable to 10 modifiable risk factors, including high blood pressure, air pollution, smoking, high LDL cholesterol, diet high in sodium, high fasting blood glucose, kidney dysfunction, excess body weight, physical inactivity, and harmful use of alcohol.

Overview

A stroke is a medical emergency that occurs when blood flow to the brain is interrupted, either due to a blockage or bleeding. This lack of blood flow can lead to brain cell death and serious complications. Strokes can be fatal and need immediate treatment.

The types of strokes are:

- ischaemic stroke: a blood clot blocks a blood vessel in your brain leading to loss of blood flow;
- haemorrhagic stroke: intracerebral haemorrhage (ICH) by rupture of a blood vessel in the brain leading to bleeding; and
- transient ischaemic attack (TIA): a short period of symptoms similar to those of a stroke, caused by a brief blockage of blood flow to the brain. A TIA usually lasts only a few minutes and doesn't cause long-term damage.

Risk factors

There are by modifiable and non-modifiable risk factors for stroke.

Modifiable risk factors are:

- hypertension: the leading contributor to stroke. Individuals with hypertension have a nearly three times higher risk of stroke compared to those without hypertension;
- tobacco use: associated with approximately two-fifths of all stroke deaths among individuals under the age of 65; and

- other risk factors: high blood cholesterol, diabetes mellitus, high levels of low-density lipoprotein cholesterol, overweight and obesity, physical inactivity, unhealthy diet (particularly diets high in sodium and low in fruits and vegetables), harmful use of alcohol, use of illicit drugs (e.g. cocaine), air pollution, high fasting blood glucose and kidney dysfunction.

Non-modifiable risk factors are:

- age
- a prior stroke
- pre-existing cardiac conditions (e.g. atrial fibrillation, heart failure), and chronic kidney disease.

Symptoms

The following symptoms can be signs of stroke:

- balance – sudden dizziness, loss of balance or coordination
- eyes – sudden vision loss
- face – facial droop
- arms – weakness in one or both arms
- speech – slurred or strange speech.

Diagnosis and treatment

When a stroke is suspected, it should be treated as a health emergency. Rapid care is essential. Immediate diagnostic imagining (a CT or MRI scan) should be carried out as soon as possible.

In the case of ischaemic stroke, thrombolytic therapy (e.g. tPA) should be carried out as soon as possible, and within hours of symptom onset. An endovascular thrombectomy may be indicated, depending on the severity and location of symptoms.

In the case of haemorrhagic stroke, management includes action to control blood pressure; intensive care in a specialized stroke unit; and possibly surgical intervention.

Admission to a specialist stroke unit with rehabilitation services enhances outcomes and decreases mortality and morbidity.

Prevention

Lifestyle changes and effective management of high blood pressure can significantly reduce the risk of stroke.

Everyone can reduce their risk of stroke by:

- controlling high blood pressure through a healthy lifestyle and appropriate antihypertensive treatment;
- reducing tobacco use and limiting exposure to second-hand smoke;
- adopting a healthy balanced diet, including at least five servings of fruit and vegetables daily and reducing intake of salt, saturated fats, and sugars;

- engaging in regular physical activity, aiming for at least 150 minutes of moderate-intensity exercise per week;
- effectively managing comorbid conditions such as diabetes, dyslipidaemia, and atrial fibrillation;
- limiting use of alcohol; and
- improving indoor and outdoor air quality.

Action can also be taken for individuals at particular risk:

- antiplatelet therapy or anticoagulants can be given, as appropriate, to prevent recurrent strokes;
- lipid-lowering therapy can be initiated for individuals at risk of atherosclerotic cardiovascular disease; and
- assessment for carotid artery stenosis can be carried out and when clinically indicated, revascularization (a medical procedure aimed at restoring the blood flow to the brain when the flow is limited or there is a blockage, as in ischemic stroke) can be considered.

Complications

Stroke can lead to a wide range of complications, including serious effects on the brain, heart and other parts of the body.

Acute complications (that appear within days to weeks) are:

- brain oedema or swelling of the brain
- aphasia or other speech disorders
- trouble swallowing (dysphagia)
- aspiration pneumonia, a lung infection
- seizure
- depression
- pressure ulcers
- limb contracture
- spasticity
- shoulder pain
- deep venous thrombosis.

Long-term complications (months to years) are:

- persistent motor deficits and gait disturbances (abnormalities in movement and walking pattern)
- chronic pain and spasticity
- cognitive impairment and memory loss

- emotional and psychological conditions, including depression and anxiety
- post-stroke epilepsy
- urinary incontinence
- loss of independence and reduced ability to perform daily activities.

Rehabilitation after stroke

Rehabilitation is a critical component of stroke care and recovery. It aims to help individuals regain functional abilities, improve quality of life, and restore independence to the greatest extent possible. Stroke rehabilitation should begin as soon as the person is medically stable – ideally within the first few days after the stroke.

Rehabilitation involves a multidisciplinary approach, including:

- physiotherapy to improve mobility, muscle strength, balance, and coordination;
- occupational therapy to enhance the ability to perform daily activities such as bathing, dressing, and cooking;
- speech and language therapy to address difficulties with communication, speech, and swallowing;
- neuropsychological and cognitive therapy to support cognitive function, memory, and emotional well-being; and
- psychological support and counselling to address depression, anxiety, and emotional adjustment after stroke.

WHO response

The World Health Organization (WHO) helps countries reduce the burden of stroke and improve prevention, treatment, and rehabilitation.

In 2024, WHO published a framework for organizing care for stroke and acute coronary syndrome. It offers guidance on timely diagnosis, effective treatment, and rehabilitation, helping countries improve services and reduce deaths and disability. WHO also supports integrating stroke care into broader health strategies and universal health coverage, ensuring services are accessible and sustainable. The WHO STEPwise approach to stroke surveillance manual helps countries collect reliable data on stroke cases and outcomes, supporting evidence-based policies.

Stroke is also part of WHO's broader work on neurological disorders. The Intersectoral global action plan on epilepsy and other neurological disorders (2022–2031) promotes better access to care and improved quality of life for people with conditions like stroke. Similarly, stroke rehabilitation care is included in the WHO Package of interventions for rehabilitation: module 3: neurological conditions.

Managing high blood pressure is key to preventing stroke. WHO's 2021 guideline for the pharmacological treatment of hypertension in adults supports health systems in treating hypertension effectively. To strengthen heart and stroke prevention, WHO launched the HEARTS initiative in 2016. It promotes healthy lifestyles, access to medicines, and team-based care.

Stroke remains a major global health challenge. Without better care, it will be difficult to reach the goal of reducing premature deaths from noncommunicable diseases by one-third by 2030 (SDG 3.4).

SOURCE-NCDC

What it is

A stroke, sometimes called a brain attack, occurs when something blocks blood supply to part of the brain or when a blood vessel in the brain bursts.

In either case, parts of the brain become damaged or die. A stroke can cause lasting brain damage, long-term disability, or even death.

Learn about the health conditions and lifestyle habits that can increase your risk for stroke.

What happens in the brain during a stroke?

The brain controls our movements, stores our memories, and is the source of our thoughts, emotions, and language. The brain also controls many functions of the body, like breathing and digestion.

To work properly, your brain needs oxygen. Your arteries deliver oxygen-rich blood to all parts of your brain. If something happens to block the flow of blood, brain cells start to die **within minutes,** because they can't get oxygen. This causes a stroke.

Learn more about the signs and symptoms of stroke.

Types

There are two types of stroke:

- Ischemic stroke.
- Hemorrhagic stroke.

A transient ischemic attack (TIA) is sometimes called a "mini-stroke." It is different from the major types of stroke. Blood flow to the brain is blocked for only a short time—usually no more than 5 minutes.1

Ischemic stroke

Most strokes are ischemic strokes. An ischemic stroke occurs when blood clots or other particles block the blood vessels to the brain.

Fatty deposits called plaque can also cause blockages by building up in the blood vessels.

Hemorrhagic stroke

A hemorrhagic stroke happens when an artery in the brain leaks blood or ruptures (breaks open). The leaked blood puts too much pressure on brain cells, which damages them.

High blood pressure and aneurysms—balloon-like bulges in an artery that can stretch and burst—are examples of conditions that can cause a hemorrhagic stroke.

Transient ischemic attack (TIA or “mini-stroke”)

TIAs are sometimes known as "warning strokes." It's important to know that:

- A TIA is a warning sign of a future stroke.

- A TIA is a medical emergency, just like a major stroke.
- Strokes and TIAs require emergency care. **Call 9-1-1** right away if you feel symptoms of a stroke or see signs in another person.
- There is no way to know in the beginning whether symptoms are from a TIA or from a major type of stroke.
- Like ischemic strokes, blood clots often cause TIAs.
- More than a third of people who have a TIA and don't get treatment have a major stroke within 1 year. As many as 10% to 15% of people will have a major stroke within 3 months of a TIA.1

Recognizing and treating TIAs can lower the risk of a major stroke. If you have a TIA, your health care team can find the cause and take steps to prevent a major stroke.

Signs and symptoms

By knowing the signs and symptoms of stroke, you can take quick action and perhaps save a life—maybe even your own.

What are the signs of stroke in men and women?

- Sudden trouble walking, dizziness, loss of balance, or lack of coordination.
- Sudden trouble seeing.
- Sudden numbness or weakness in the face, arm, or leg, especially on one side of the body.
- Sudden confusion, trouble speaking, or difficulty understanding speech.
- Sudden severe headache with no known cause.

Call 9-1-1 right away if you or someone else has any of these symptoms.

B.E. F.A.S.T to identify stroke

B.E. F.A.S.T. to help stroke patients get the treatments they need.

The stroke treatments that work best are available only if the stroke is recognized and diagnosed within 3 hours of the first symptoms. Stroke patients may not be eligible for these treatments if they don't arrive at the hospital in time.

If you think someone may be having a stroke, B.E. F.A.S.T. and do the following test:

- **B—Balance Loss:** Ask the person if they are feeling off-balance or dizzy.
- **E—Eye (Vision) Changes:** Ask the person if they have trouble seeing normally.
- **F—Face:** Ask the person to smile. Does one side of the face droop?
- **A—Arms:** Ask the person to raise both arms. Does one arm drift downward?
- **S—Speech:** Ask the person to repeat a simple phrase. Is the speech slurred or strange?
- **T—Time:** If you see any of these signs, call 9-1-1 right away.

Note the time when any symptoms first appear. This information helps health care providers determine the best treatment.

Do not drive to the hospital or let someone else drive you. Call 9-1-1 for an ambulance so that medical personnel can begin life-saving treatment on the way to the emergency room.

What should I do to treat a transient ischemic attack ("mini-stroke")?

If your stroke symptoms go away after a few minutes, you may have had a transient ischemic attack (TIA), also sometimes called a "mini-stroke." Although brief, a TIA is a sign of a serious condition that will not go away without medical help.

Unfortunately, because TIAs clear up, many people ignore them. But paying attention to a TIA can save your life. If you think you or someone you know has had a TIA, tell a health care team about the symptoms right away.

How is stroke treated?

On the way to the hospital

Calling 9-1-1 for an ambulance means that medical staff can begin life-saving treatment on the way to the emergency room.

Do not drive to the hospital or let someone else drive you. The key to stroke treatment and recovery is getting to the hospital quickly.

Stroke patients who are taken to the hospital in an ambulance may get diagnosed and treated more quickly than people who do not arrive in an ambulance.1 This is because emergency treatment starts on the way to the hospital. The emergency workers may take you to a specialized stroke center to ensure that you receive the quickest possible diagnosis and treatment.

The emergency workers will also collect valuable information that guides treatment and alert hospital medical staff before you arrive at the emergency room, giving them time to prepare.

Prevention steps and strategies

Choose healthy foods and drinks

Choosing healthy meal and snack options can help you prevent stroke. Be sure to eat plenty of fresh fruits and vegetables.

Eating foods low in saturated fats, trans fat, and cholesterol and high in fiber can help prevent high cholesterol. Limiting salt (sodium) in your diet can also lower your blood pressure. High cholesterol and high blood pressure increase your chances of having a stroke.

For more information on healthy diet and nutrition, see CDC's Nutrition website.

Keep a healthy weight

Having overweight or obesity increases your risk for stroke. To determine whether your weight is in a healthy range, doctors often calculate your body mass index (BMI). If you know your weight and height, you can calculate your BMI at CDC's BMI calculator. Doctors sometimes also use waist and hip measurements to measure excess body fat.

Get regular physical activity

Physical activity can help you stay at a healthy weight and lower your cholesterol and blood pressure levels. For adults, the surgeon general recommends 2 hours and 30 minutes of moderate-intensity aerobic physical activity, such as a brisk walk, each week. Children and teens should get 1 hour of physical activity every day.

Find inspiration and tips for getting regular physical activity from the "Live to the Beat" campaign.

Don't smoke

Cigarette smoking greatly increases your chances of having a stroke. If you don't smoke, don't start. If you do smoke, quitting will lower your risk for stroke. Your doctor can suggest ways to help you quit.

For more information about tobacco use and quitting, see CDC's Smoking & Tobacco Use website.

Limit alcohol

Avoid drinking too much alcohol, which can raise your blood pressure. Men should have no more than two drinks per day, and women should have no more than one per day. For more information, visit CDC's Alcohol and Public Health website.

Control your medical conditions

Talk with your doctor about steps you can take to lower your risk for stroke.

If you have heart disease, high cholesterol, high blood pressure, or diabetes, you can take steps to lower your risk for stroke.

Check cholesterol

Your doctor should test your cholesterol levels at least once every 5 years. Talk with your health care team about this simple blood test. If you have high cholesterol, medicine and lifestyle changes can help lower your risk for stroke.

Control blood pressure

High blood pressure usually has no symptoms, so be sure to have it checked on a regular basis. Talk to your health care team about how often you should check your levels. You can check your blood pressure at home, at a doctor's office, or at a pharmacy.

If you have high blood pressure, your doctor might prescribe medicine, suggest some changes in your lifestyle, or recommend that you choose foods with lower sodium (salt).

Control diabetes

If your doctor thinks you have symptoms of diabetes, he or she may recommend that you get tested. If you have diabetes, check your blood sugar levels regularly.

Talk with your health care team about treatment options. Your doctor may recommend certain lifestyle changes, such as getting more physical activity or choosing healthier foods. These actions will help keep your blood sugar under control and help lower your risk for stroke.

Treat heart disease

If you have certain heart conditions, such as coronary artery disease or atrial fibrillation (irregular heartbeat), your health care team may recommend medical treatment or surgery. Taking care of heart problems can help prevent stroke.

Take your medicine

If you take medicine to treat heart disease, high cholesterol, high blood pressure, or diabetes, follow your doctor's instructions carefully. Always ask questions if you don't understand something. Never stop taking your medicine without first talking to your doctor or pharmacist.

Work with your health care team

You and your health care team can work together to prevent or treat the medical conditions that lead to stroke. Discuss your treatment plan regularly, and bring a list of questions to your appointments. Learn how to find the right doctor for you from the "Live to the Beat" campaign.

If you've already had a stroke or transient ischemic attack (TIA), your health care team will work with you to prevent further strokes. Your treatment plan will include medicine or surgery and lifestyle changes to lower your risk for another stroke. Be sure to take your medicine as directed and follow your doctor's instructions.